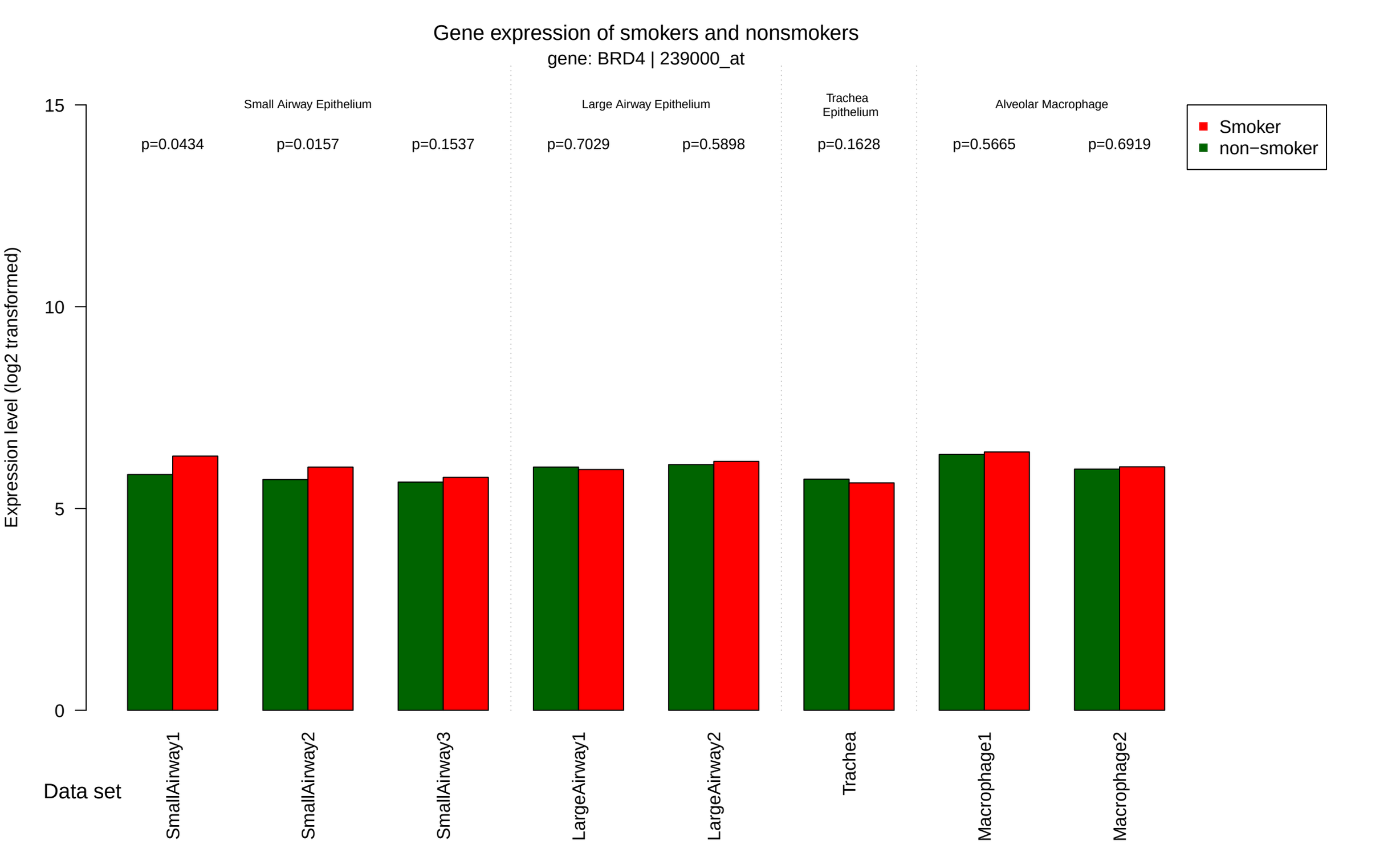
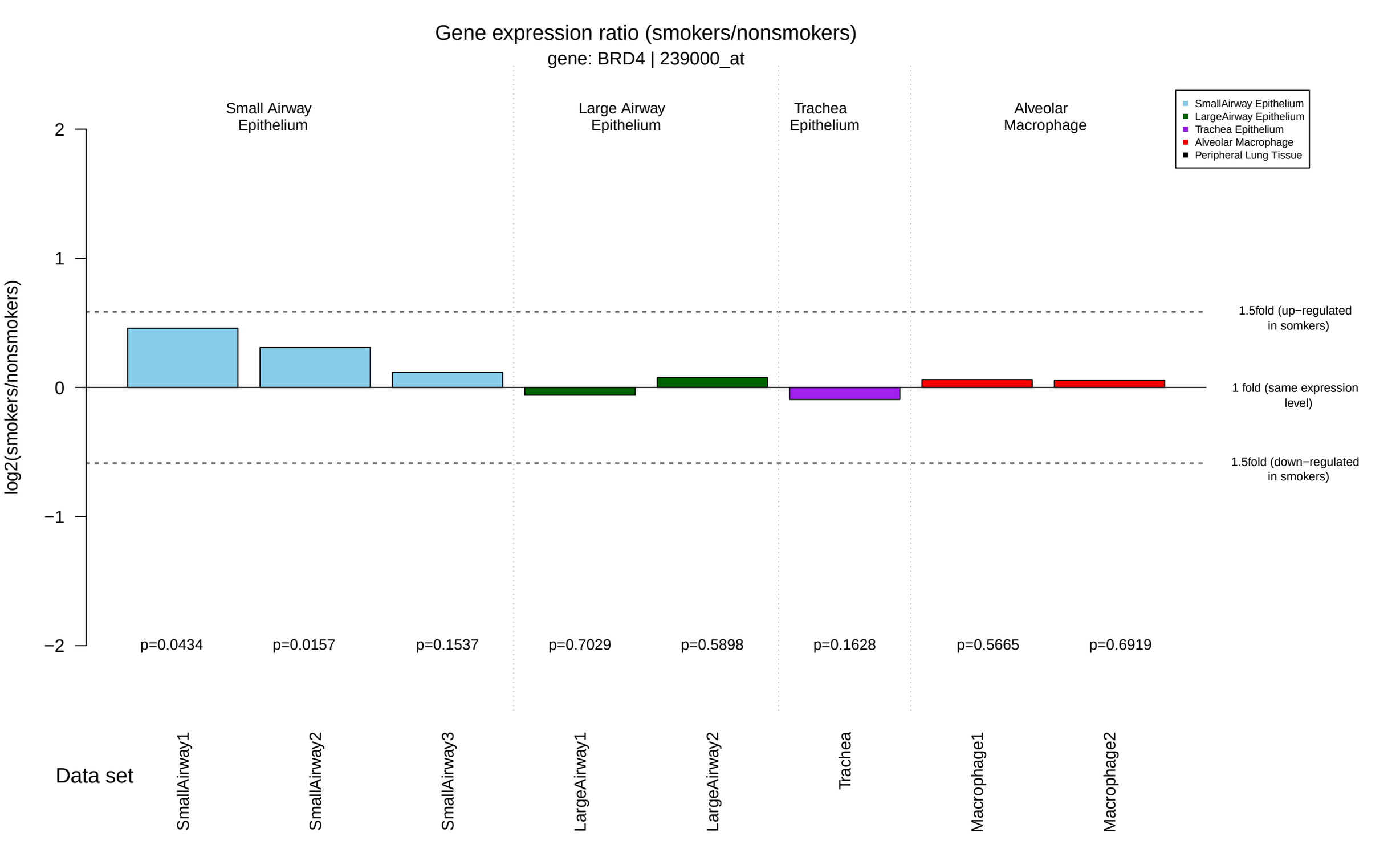
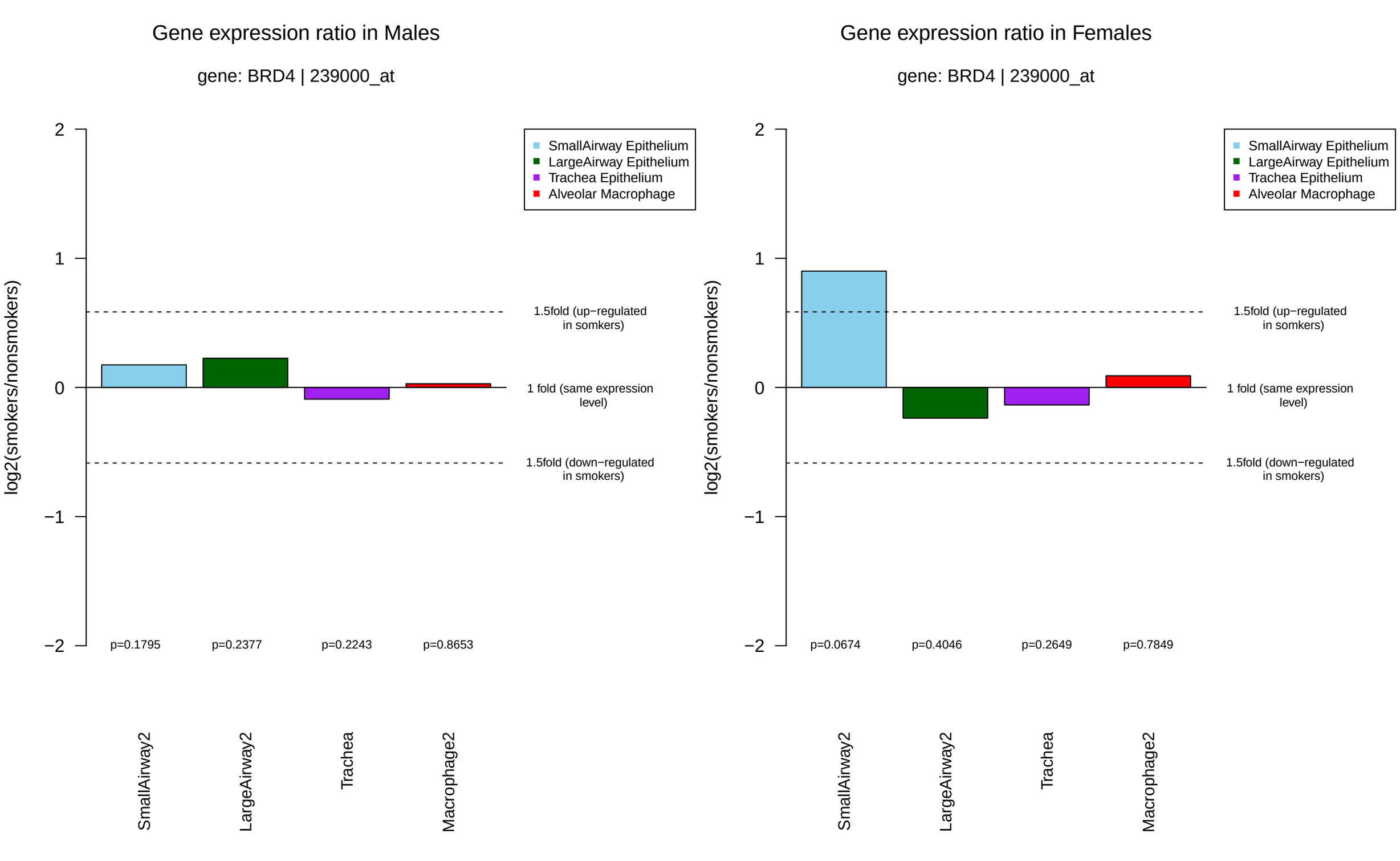
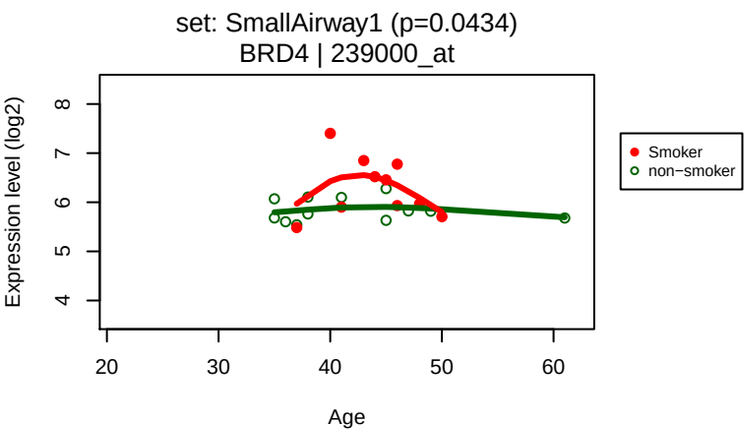
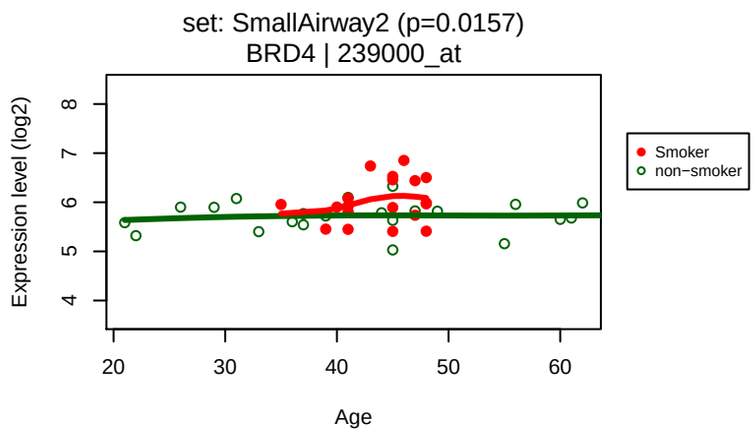
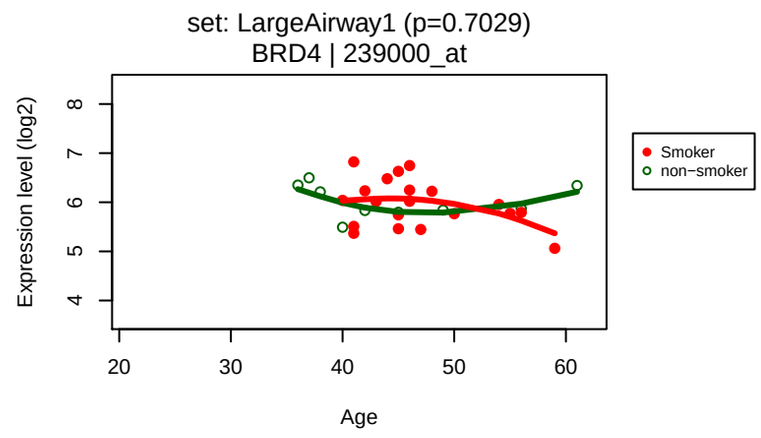

Gene expression of smokers and nonsmokers
gene: BRD4 | 239000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0434
p=0.0157
p=0.1537
p=0.7029
p=0.5898
p=0.1628
p=0.5665
p=0.6919
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

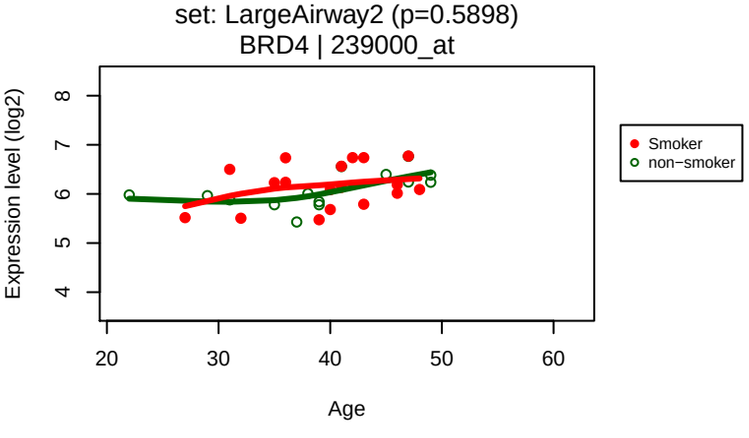

Gene expression ratio (smokers/nonsmokers)
gene: BRD4 | 239000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0434
p=0.0157
p=0.1537
p=0.7029
p=0.5898
p=0.1628
p=0.5665
p=0.6919
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

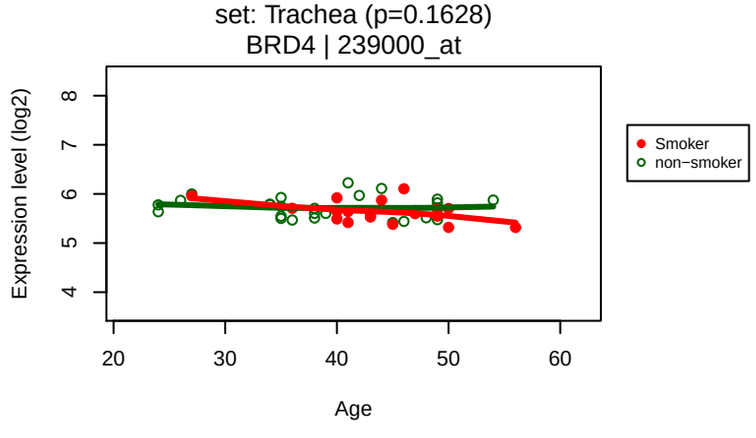

Gene expression ratio in Males
gene: BRD4 | 239000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1795
p=0.2377
p=0.2243
p=0.8653
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRD4 | 239000_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0674
p=0.4046
p=0.2649
p=0.7849
SmallAirway2
LargeAirway2
Trachea
Macrophage2

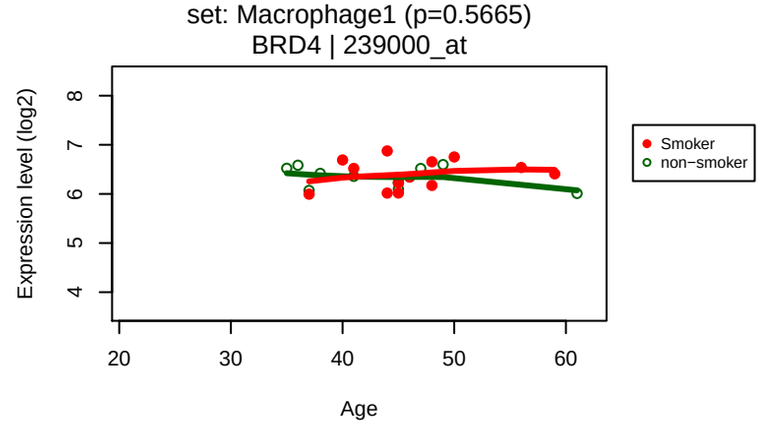

set: SmallAirway1 (p=0.0434)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker

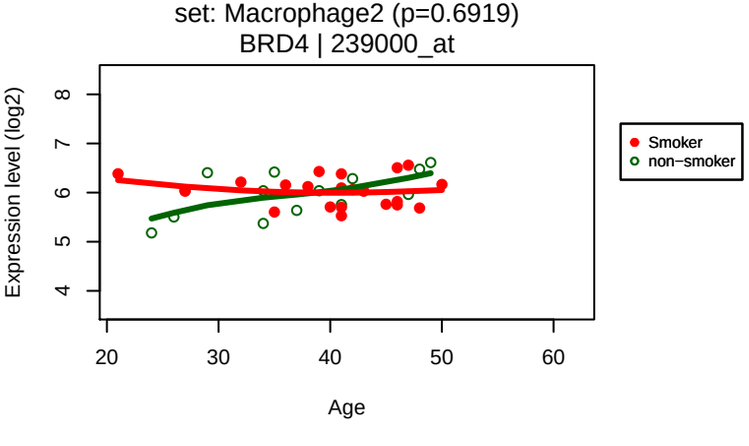

set: SmallAirway2 (p=0.0157)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker

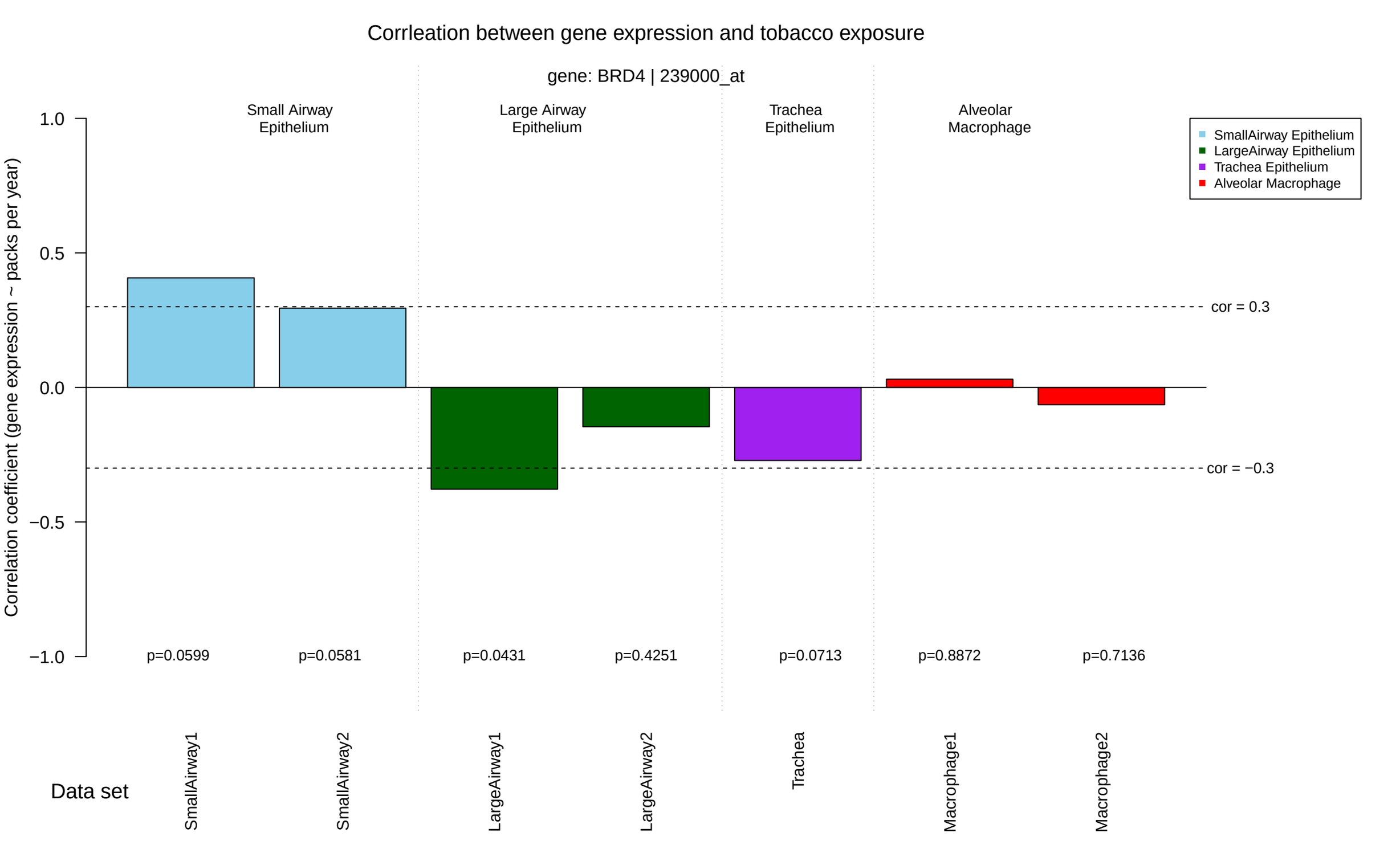

set: LargeAirway1 (p=0.7029)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5898)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1628)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5665)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6919)
BRD4 | 239000_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRD4 | 239000_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0599
p=0.0581
p=0.0431
p=0.4251
p=0.0713
p=0.8872
p=0.7136
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2